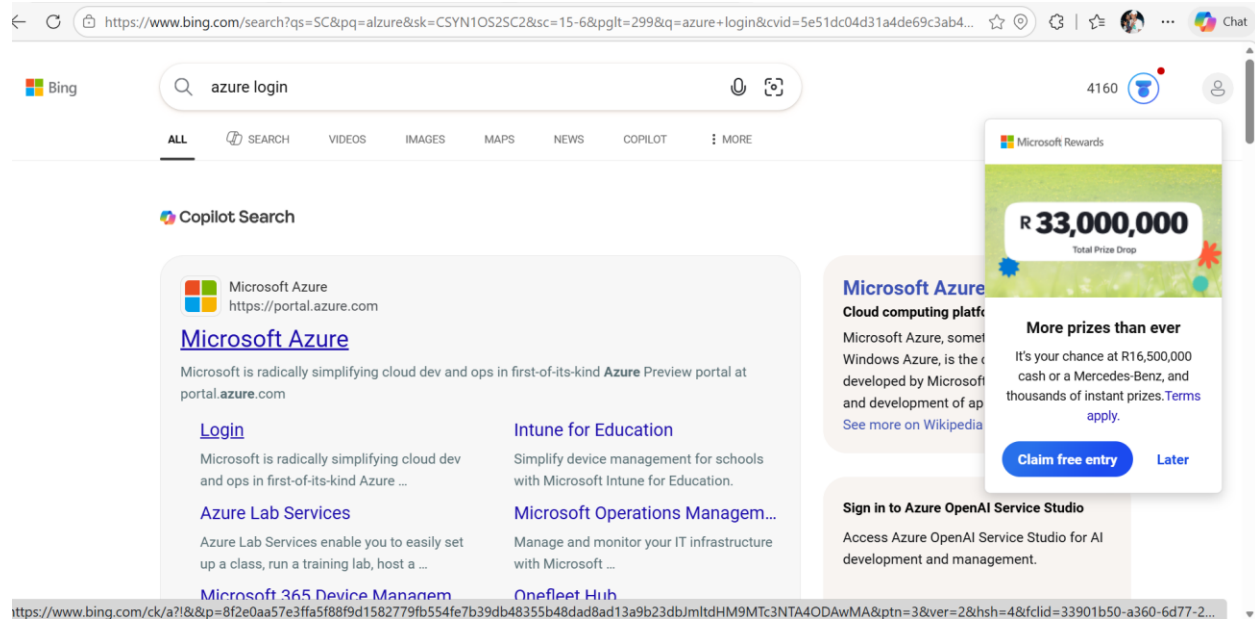


## Chapter 1 Lab: Introduction to Azure & Resource Management

### Task1: Login to Azure Portal

Use any search engine, such as Google or Bing (as I did) to access Microsoft Azure (<https://portal.azure.com>).



I used a personal Microsoft account to access the Azure portal because my university email address was not permitted to sign in. The default “Azure subscription 1” was used for all lab activities. Since the lab is for learning purposes, the available subscription provided sufficient access to create and manage resources.

### Task 2: Create a Resource Group

Microsoft Azure portal home page. The top navigation bar includes the Microsoft Azure logo, a search bar, and a Copilot button. The main content area is divided into two sections: "Azure services" and "Resources".

**Azure services**

- Create a resource
- Resource groups
- Automation Accounts
- Virtual machines
- App Services
- All resources
- Quickstart Center
- Foundry
- Kubernetes services
- More services

**Resources**

Recent Favorite

| Name                                                      | Type                    | Last Viewed |
|-----------------------------------------------------------|-------------------------|-------------|
| Function                                                  | Function App            | an hour ago |
| rgcomputeab885                                            | Storage account         | 2 hours ago |
| ASP-RGComputeLab-a31e                                     | App Service plan        | 2 hours ago |
| Function                                                  | Application Insights    | 2 hours ago |
| RG-ComputeLab                                             | Resource group          | 2 hours ago |
| app-Ntibo                                                 | App Service             | 3 hours ago |
| DefaultWorkspace-bfe2eef6-3077-4a69-bc00-11e40eef69df-IN8 | Log Analytics workspace | 3 hours ago |
| app-Ntibo                                                 | Application Insights    | 3 hours ago |
| APS-RGComputeLab-9175                                     | App Service plan        | 3 hours ago |
| VM-Linux-Ntibo                                            | Virtual machine         | 4 hours ago |
| VM-Linux-Ntibo-ip                                         | Public IP address       | 4 hours ago |
| VM-Linux-Ntibo-vnet                                       | Virtual network         | 4 hours ago |

See all

Microsoft Azure portal "Create a resource" page. The top navigation bar includes the Microsoft Azure logo, a search bar, and a Copilot button. The main content area is divided into two sections: "Get Started" and "Categories".

**Create a resource**

Design a new Azure workload Help me choose the right VM size I want to duplicate an existing VM

**Get Started**

Recently created

**Categories**

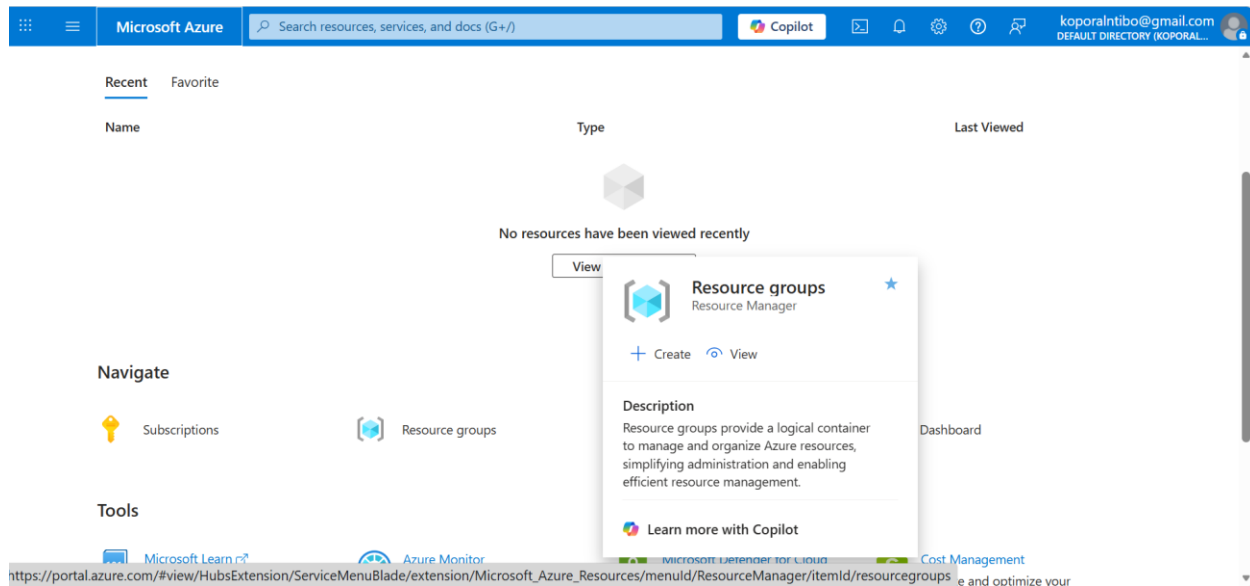
- Machine Learning
- AI Apps and Agents
- Analytics
- Blockchain
- Infrastructure Services
- Databases
- Developer Tools
- DevOps
- Identity
- Integration

**Search results for "resource group"**

- resource group
- resource groups
- Move the Chain Engage
- Function App Create
- Logic App Create
- Web App Create
- Virtual network Create
- Resource group Create

Getting started? Try our Quickstart Center

Give feedback



When logged in, use search bar and search for “Resource Groups” and “Create”. Resource groups are used to separate Testing, Development, and Production. As they help logically organize projects, assign access easily, and apply policies easily.

Home

## Create a resource group

Basics Tags Review + create

**Resource group** - A container that holds related resources for an Azure solution. The resource group can include all the resources for the solution, or only those resources that you want to manage as a group. You decide how you want to allocate resources to resource groups based on what makes the most sense for your organization. [Learn more](#)

Subscription \* ⓘ Azure subscription 1

Resource group name \* ⓘ SM\_Ntibo

Region \* ⓘ (Africa) South Africa North

Previous Next Review + create

- I used the default **“Azure subscription 1”** because my university email did not allow access to the Azure for Students subscription. However, it still provided the necessary features to complete the lab tasks.
- I selected the region **“South Africa North”** since it is closest to my location, which helps improve performance and reduce latency.

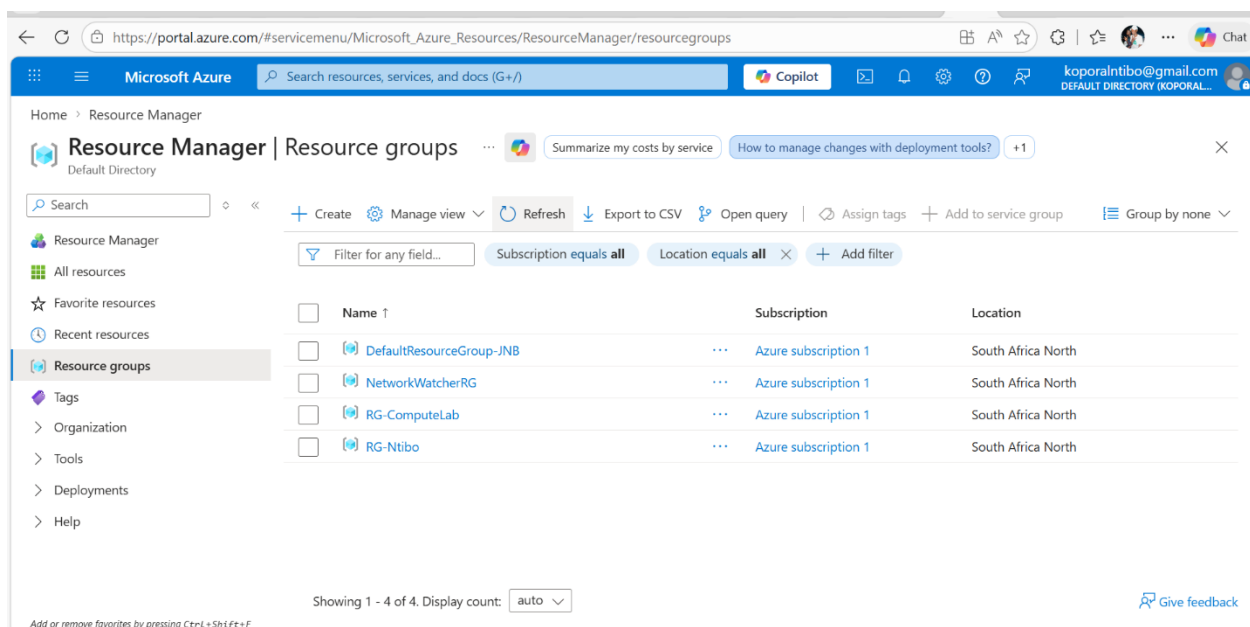
After opening the **“Create a Resource Group”** section, you must select a subscription. In this case, the selected subscription was **“Azure subscription 1.”**

A subscription determines:

- The billing boundary
- Who is responsible for payment
- Who has access and permissions

When creating the resource group, it is important to assign a clear and meaningful name. This helps in identifying resources easily and avoids confusion when managing multiple projects.

Next, choose the location of the resource group, which refers to the physical Azure data center (region). Selecting the correct region is important for latency, legal compliance, and cost efficiency. The region used here is **South Africa North**, as it is geographically closer.

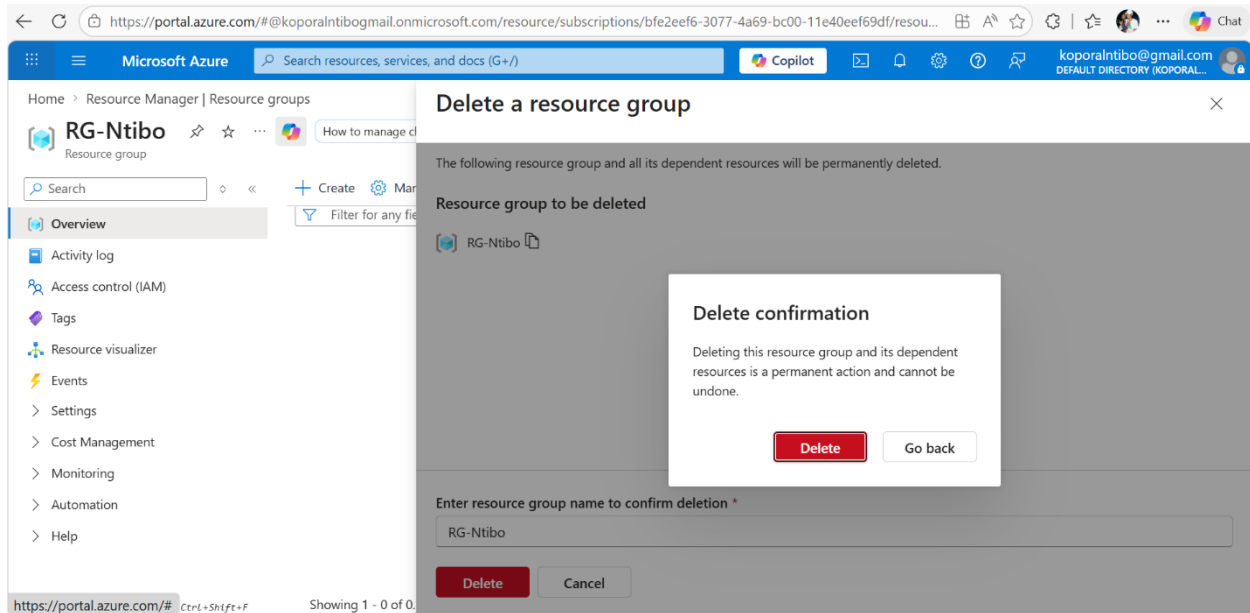
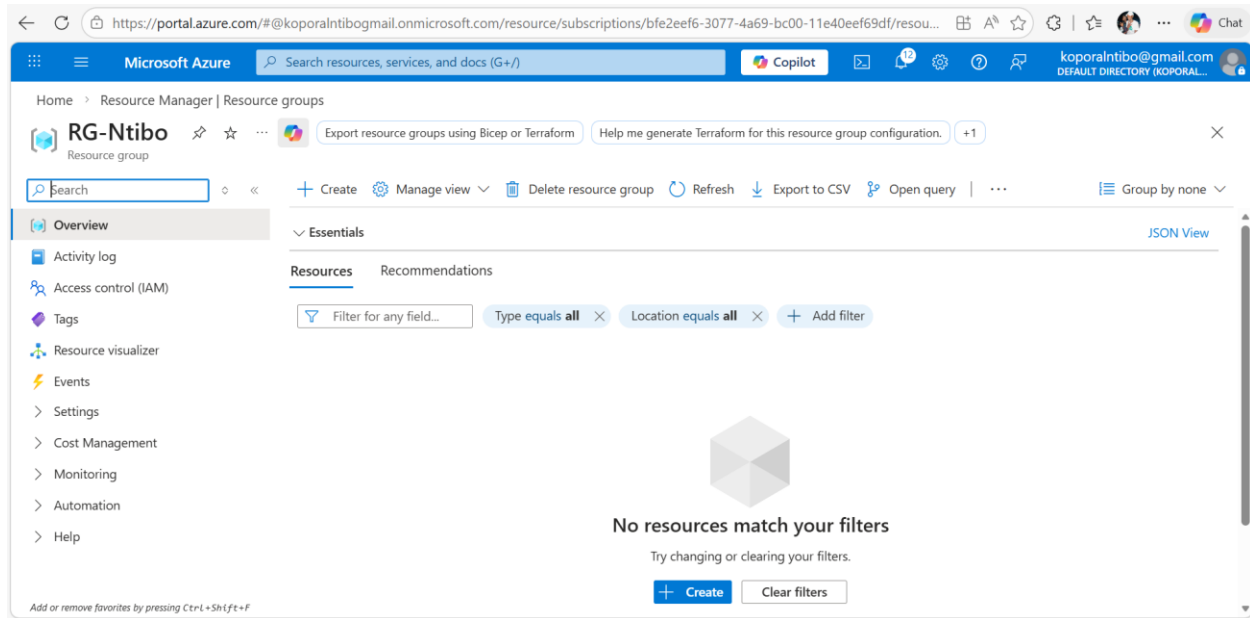


The screenshot shows the Azure portal interface for the Resource Manager. The left sidebar contains navigation options: Home, Resource Manager, All resources, Favorite resources, Recent resources, Resource groups (selected), Tags, Organization, Tools, Deployments, and Help. The main content area displays a table of resource groups under the heading 'Resource Manager | Resource groups'. The table has columns for Name, Subscription, and Location. Four resource groups are listed, all under 'Azure subscription 1' and in the 'South Africa North' location.

| Name                     | Subscription         | Location           |
|--------------------------|----------------------|--------------------|
| DefaultResourceGroup-JNB | Azure subscription 1 | South Africa North |
| NetworkWatcherRG         | Azure subscription 1 | South Africa North |
| RG-ComputeLab            | Azure subscription 1 | South Africa North |
| RG-Ntibo                 | Azure subscription 1 | South Africa North |

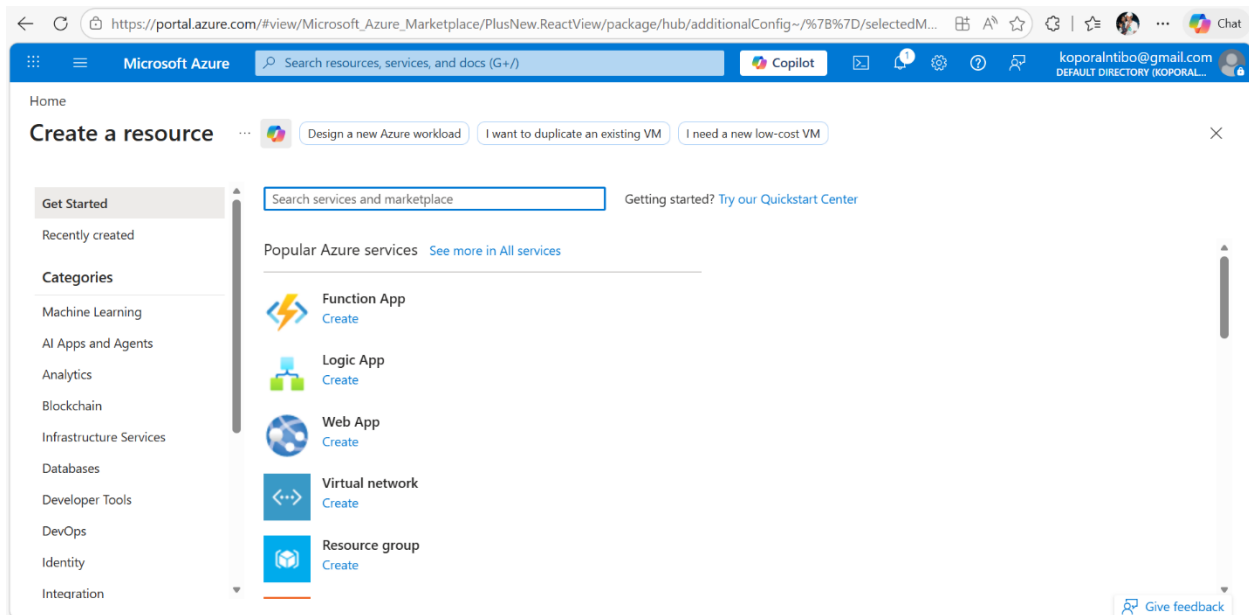
At the bottom of the table, it says 'Showing 1 - 4 of 4. Display count: auto'. There is also a 'Give feedback' link in the bottom right corner.

After the creation process is completed, the resource group will appear under **Resource Manager**.

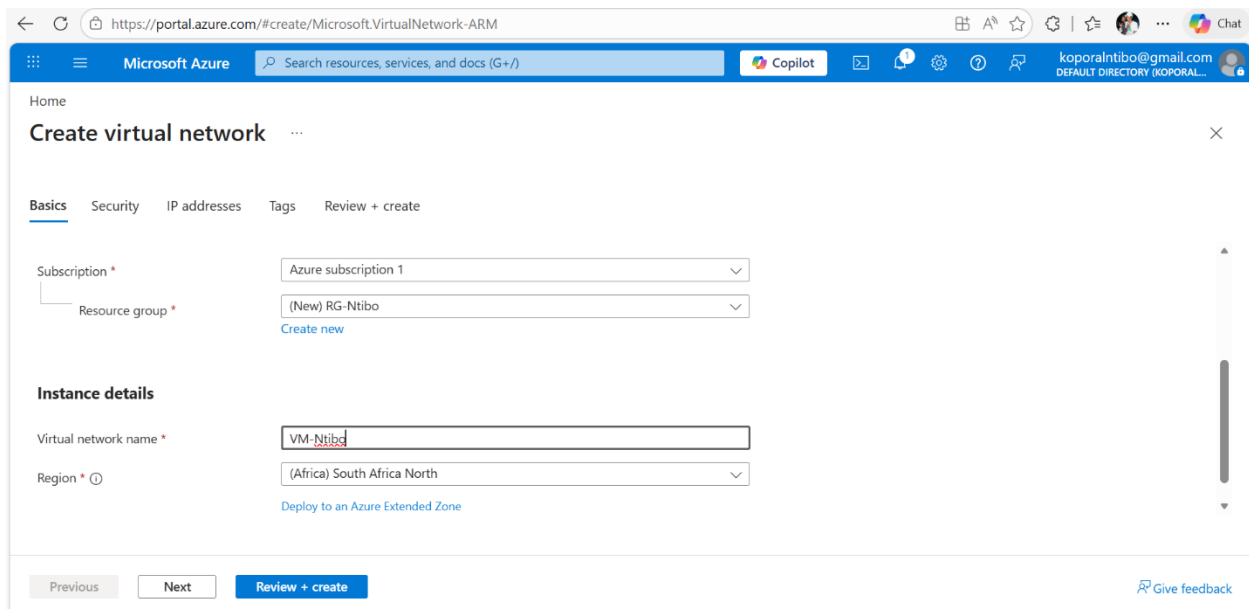


### Task 3: Deploy a Virtual Machine (IaaS)

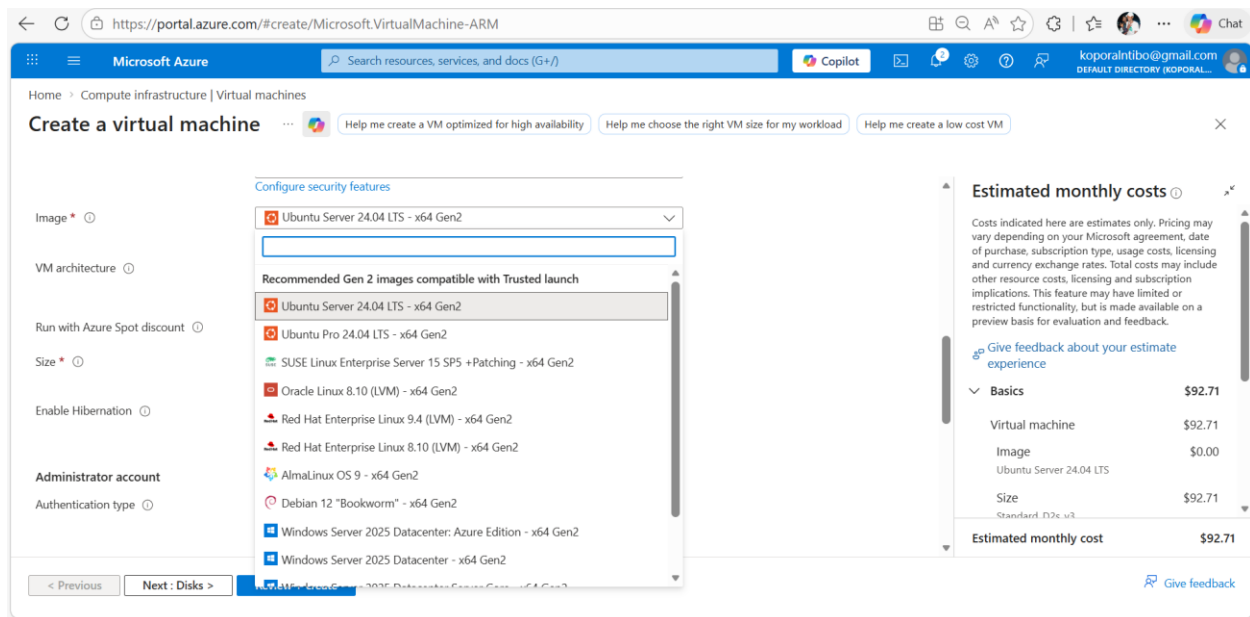




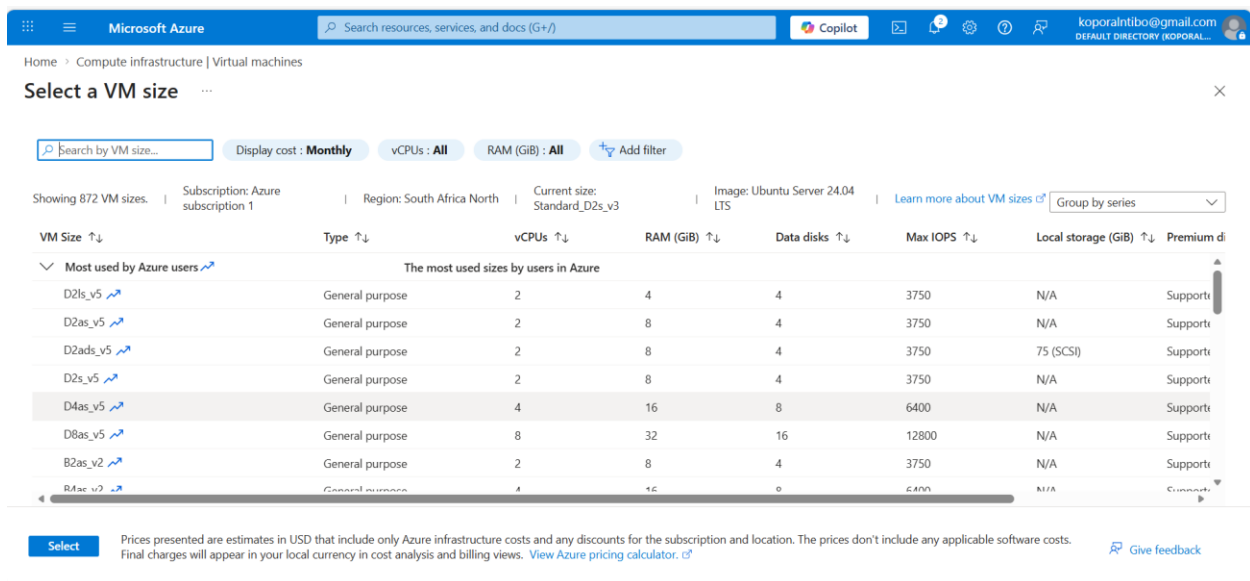
Enter the “Resource group” framework and use the search bar within to search for Virtual Machine, then click create.



When in the “Create virtual machine” framework, fill in all required information such as your subscription type, resource group name, virtual machine name and region selection. Select same subscription as when creating the resource group. Select resource group name (Resources must belong to a group). Name your Virtual Machine to distinguish among others you may create. Select the same region as your resource group (keep related resources in the same region).



Select the type of image to use for your Virtual Machine. Prefer the use of Windows Server 2025 Datacentre: Azure Edition -x64 GenZ, as it provides a fully managed optimized OS with native integration for hybrid benefits, enhance security (hotpatching), and seamless compatibility with Microsoft technologies. Hotpatching – applying security updates without needing to reboot, ensuring higher availability. Image = Operating System template.



You can also select VM size which defines memory, storage, and networking capacity for virtual machines, categorized by workload type. I have selected B2as\_v2 as it is one of the most used sizes in Azure, also it is also cost effective than most sizes, yet still effective, as

it is ideal for low-to-moderate baseline workloads that occasionally require high CPU performance, such as development servers, small databases, and web servers.

Home > Create a resource > Create a virtual machine

## Select a VM size

Minimise

Search by VM size... vCPUs: All RAM (GiB): All Display cost: Monthly Add filter

Showing 872 VM sizes. | Subscription: Azure for Students | Region: South Africa North | Current size: Standard\_B2as\_v2 | Image: Windows Server 2025 Datacenter: Azure Edition | [Learn more about VM sizes](#) Group by series

| VM Size ↑↓                                                           | Type ↑↓         | vCPUs ↑↓ | RAM (GiB) ↑↓ | Data disks ↑↓ | Max IOPS ↑↓ | Local storage (GiB) ↑↓ | Premium disk ↑↓ | Cost/month ↑↓ |
|----------------------------------------------------------------------|-----------------|----------|--------------|---------------|-------------|------------------------|-----------------|---------------|
| Most used by Azure users ↗                                           |                 |          |              |               |             |                        |                 |               |
| The most used sizes by users in Azure                                |                 |          |              |               |             |                        |                 |               |
| B2as_v2 ↗                                                            | General purpose | 2        | 8            | 4             | 3750        | N/A                    | Supported       | US\$78.11     |
| B4as_v2 ↗                                                            | General purpose | 4        | 16           | 8             | 6400        | N/A                    | Supported       | US\$156.22    |
| B-Series v2                                                          |                 |          |              |               |             |                        |                 |               |
| Ideal for workloads that do not need continuous full CPU performance |                 |          |              |               |             |                        |                 |               |
| B2as_v2 ↗                                                            | General purpose | 2        | 8            | 4             | 3750        | N/A                    | Supported       | US\$78.11     |
| B4as_v2 ↗                                                            | General purpose | 4        | 16           | 8             | 6400        | N/A                    | Supported       | US\$156.22    |
| B2als_v2                                                             | General purpose | 2        | 4            | 4             | 3750        | N/A                    | Supported       | US\$42.34     |
| B2ats_v2 (free services eligible) ⓘ                                  | General purpose | 2        | 1            | 4             | 3750        | N/A                    | Supported       | US\$15.62     |
| B2ls_v2                                                              | General purpose | 2        | 4            | 4             | 3750        | N/A                    | Supported       | US\$46.36     |
| B2s_v2                                                               | General purpose | 2        | 8            | 4             | 3750        | N/A                    | Supported       | US\$86.14     |

Home > Create a resource

## Create a virtual machine

Help me create a VM optimized for high availability Help me choose the right VM size for my workload Help me create a low cost VM

Minimise

VM architecture

x64

Arm64 is not supported with the selected image.

Run with Azure Spot discount ⓘ ☐

Size \* ⓘ 

Standard\_B2ats\_v2 - 2 vcpus, 1 GiB memory (US\$15.62) (free services eligible) ▼

See all sizes

Enable Hibernation ⓘ ☐

Hibernate is not supported by the size that you have selected. Choose a size that is compatible with Hibernation to enable this feature. [Learn more](#) ⓘ

Administrator account

Username \* ⓘ 

studentadmin ✓

Password \* 

\*\*\*\*\* ✓

Confirm password \* 

\*\*\*\*\* ✓

Inbound port rules

Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular

< Previous Next: Disks > Review + create

You can also create an Administrator account, which will only allow the admin to access the VM's privilege mode for configurations. Use the suggested username "studentadmin", and your own strong password. Also setup to allow for RDP (3389) (Remote Desktop Protocol) port only to access your VM.



#### Administrator account

Username \* ⓘ  ✓

Password \*  ✓

Confirm password \*  ✓

#### Inbound port rules

Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.

Public inbound ports \* ⓘ

☐ None

☒ Allow selected ports

Select inbound ports \*  ✓

 **This will allow all IP addresses to access your virtual machine.** This is only recommended for testing. Use the Advanced controls in the Networking tab to create rules to limit inbound traffic to known IP addresses.

Virtual Machine Creation and deployment complete, meaning the VM is fully running and operational.`

# Resource Manager | Resource groups

Default Directory



[+ Create](#) [Manage view](#) [...](#)

Resource Manager

All resources

Favorite resources

Recent resources

Resource groups

Tags

> Organization

> Tools

> Deployments

> Help

☐ Name ↑

☐ DefaultResourceGroup-JNB

☐ NetworkWatcherRG

☐ RG-ComputeLab

☐ RG-Ntibo

☐ VM-Ntibo\_group

Task 4: Apply tags

← ↻ https://portal.azure.com/#view/HubsExtension/BrowseResourcesWithTag/tagName/Lab%3AComputing/tagValue/

Microsoft Azure studentadmin Copilot kpopalntibco@gmail.com

Home > Resource Manager | Tags

Resource Manager | Tags

Search Refresh Feedback

Resource Manager

All resources

Favorite resources

Recent resources

Resource groups

Tags

Organization

Tools

Deployments

Help

Tags are name/value pairs that enable you to categorize resources and view consolidated billing by applying the same tag to multiple resources and resource groups. Tag names are case insensitive, but tag values are case sensitive. [Learn more about tags](#)

Filter items...

Tags ↑↓

Lab:Computing

Resources with tag Lab:Computing

Default Directory (kpopalntibco@gmail.onmicrosoft.com)

Manage view Refresh Export to CSV Open query Assign tags

Filter for any field... Subscription equals all Type equals all Add filter

0 Unsecure resources 0 Recommendations 0 Changed resources

No grouping List view

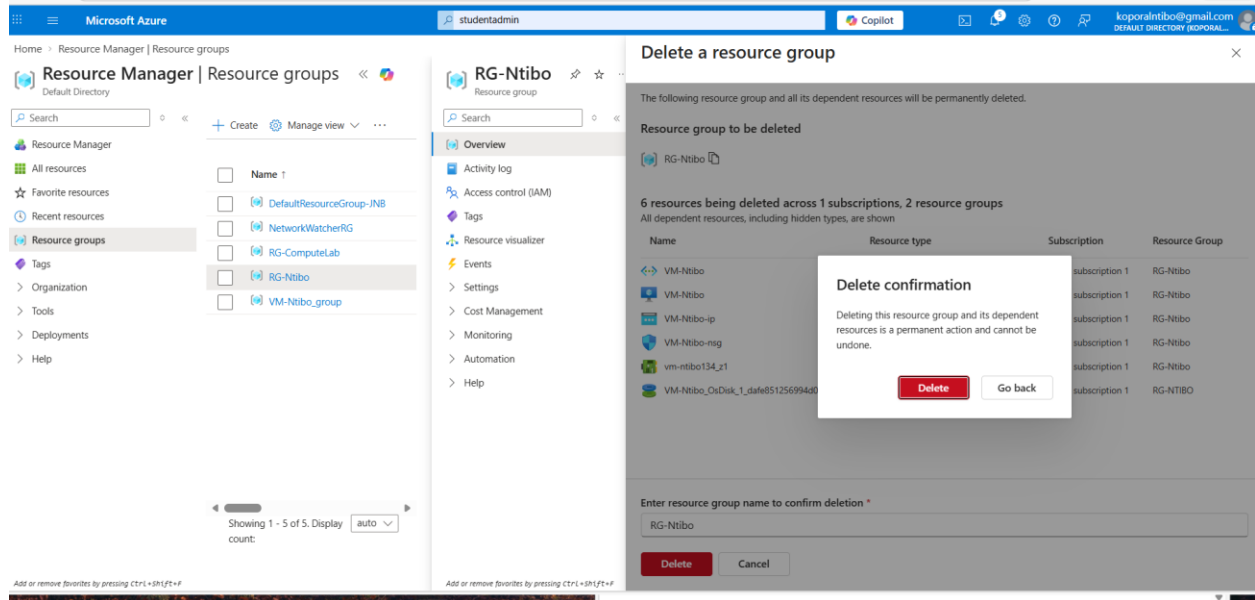
| Name ↑↓          | Type ↑↓        | Resource group ↑↓ | Location ↑↓        | Subscription ↑↓      |
|------------------|----------------|-------------------|--------------------|----------------------|
| NetworkWatcherRG | Resource group | NetworkWatcherRG  | South Africa North | Azure subscription 1 |

< Previous Page 1 of 1 Next > Showing 1 to 1 of 1 records

Give feedback

Applying Tags in your VM is crucial for effective cost management, resource organization, security, and automation. Tags are metadata, in form of key: value pairs, that help identify resources based on criteria relevant to an organization or group.

## Task 5: Delete Resource Group



Deleting resources groups in essential for cost optimization, security, and operational efficiency. Deleting a resource group, deletes everything inside it. It is efficient and clean.